

SUBMITTED BY:

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SECTION: A

SESSION:2023-2027

SUBMITTED TO:

INSTRUCTOR: RESPECTED Mr. WAQAS SALEEM

COURSE TITLE: DEVOPS

SUBMISSION DATE:

OCTOBER 06, 2026

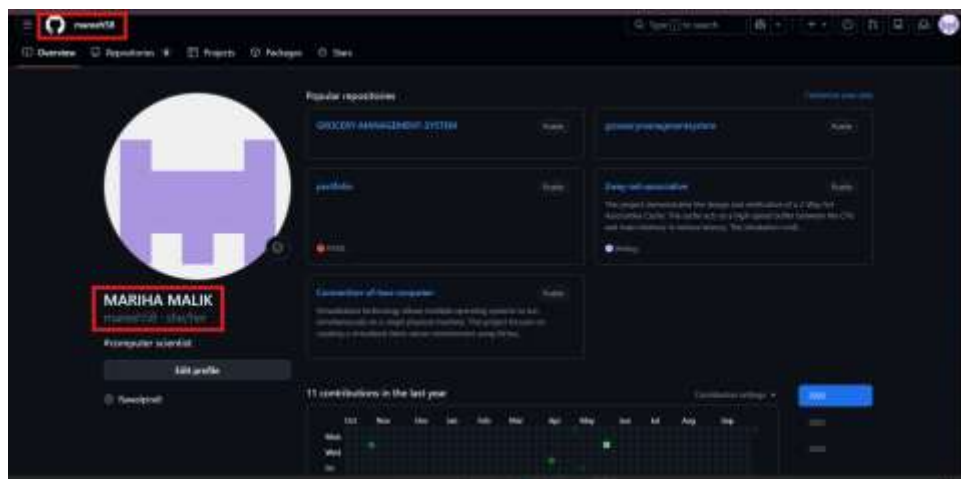
1. Introduction:

The objective of this lab is to configure an isolated DevOps virtualization environment, install Ubuntu Linux, establish basic network connectivity, configure OpenSSH server, and verify remote shell access from the host operating system.

2. Lab Execution & Task Breakdown:

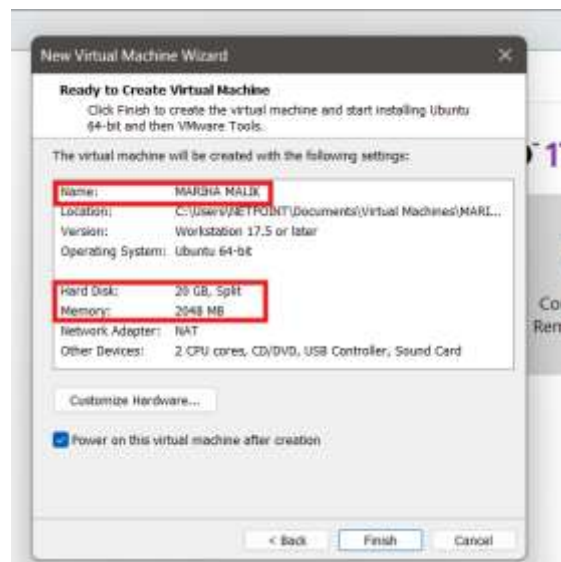
2.1. Task 1 & 2: GitHub Setup and Student Marks Portal

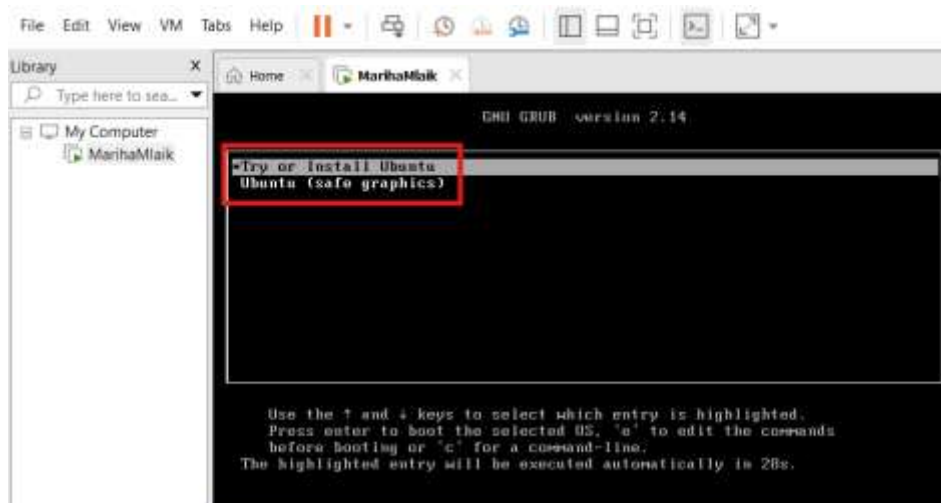
Configured personal GitHub account credentials and verified integration with the course portal for automated evaluation.



2.2. Task 3 & 4: Virtual Machine Creation and ISO Boot

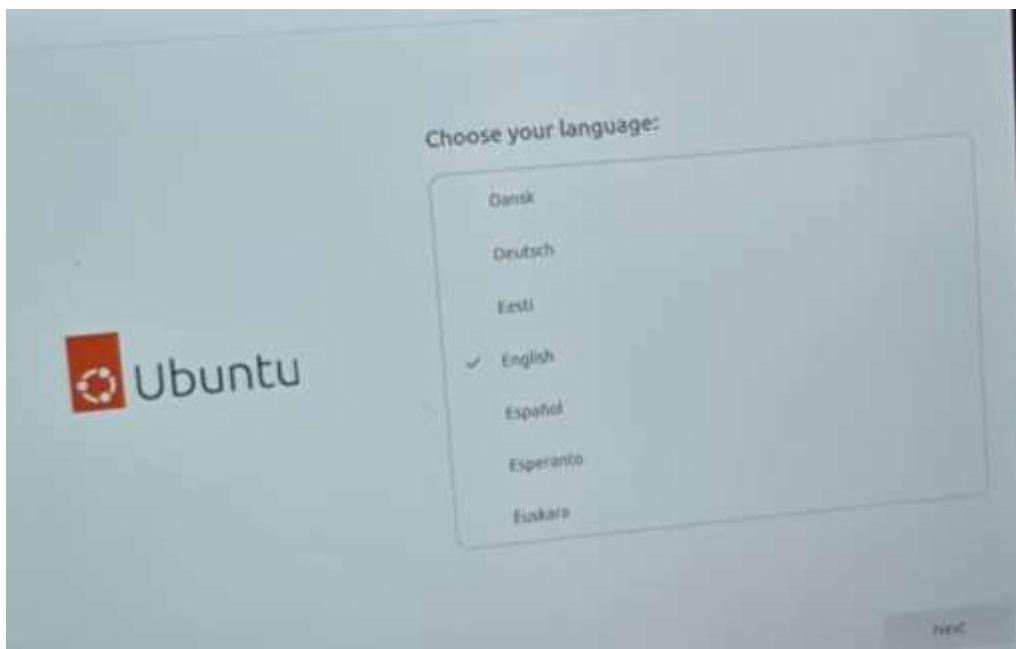
Provisioned a virtual machine with appropriate hardware specifications and initiated system boot using the Ubuntu installation media.

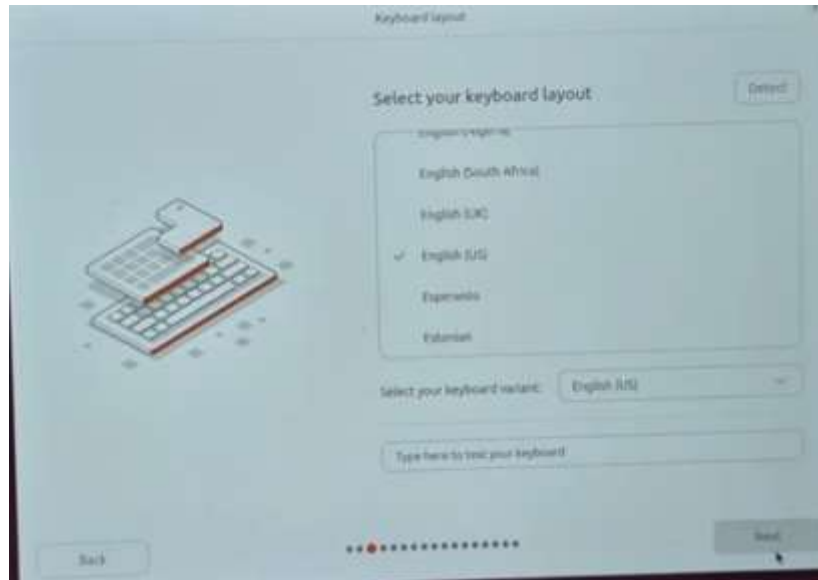




2.3. Task 5 & 6: Language and Localization

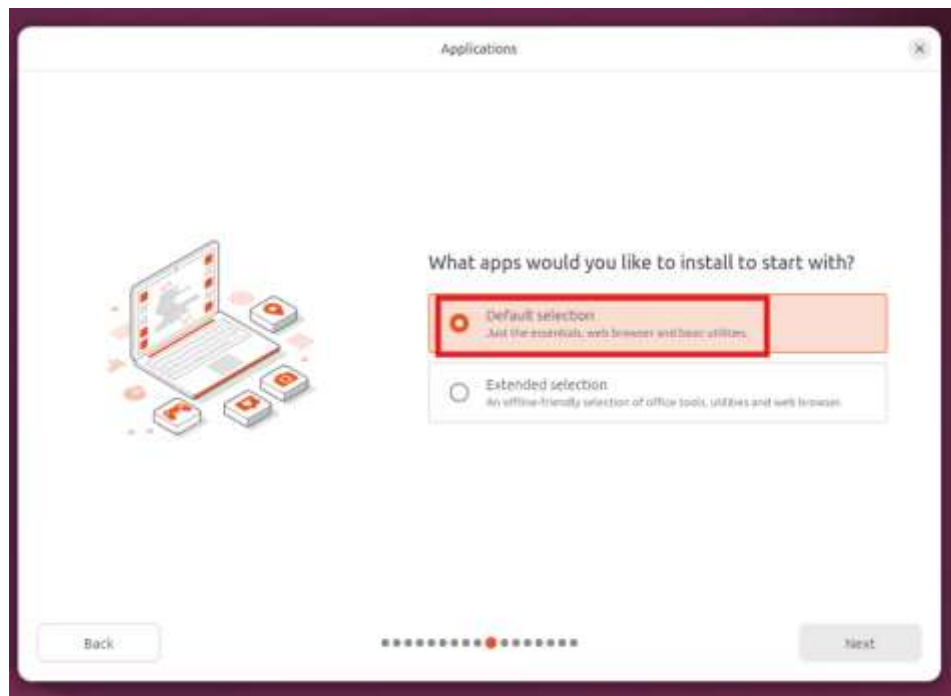
Selected English as the primary installation language and verified default standard keyboard layout settings.





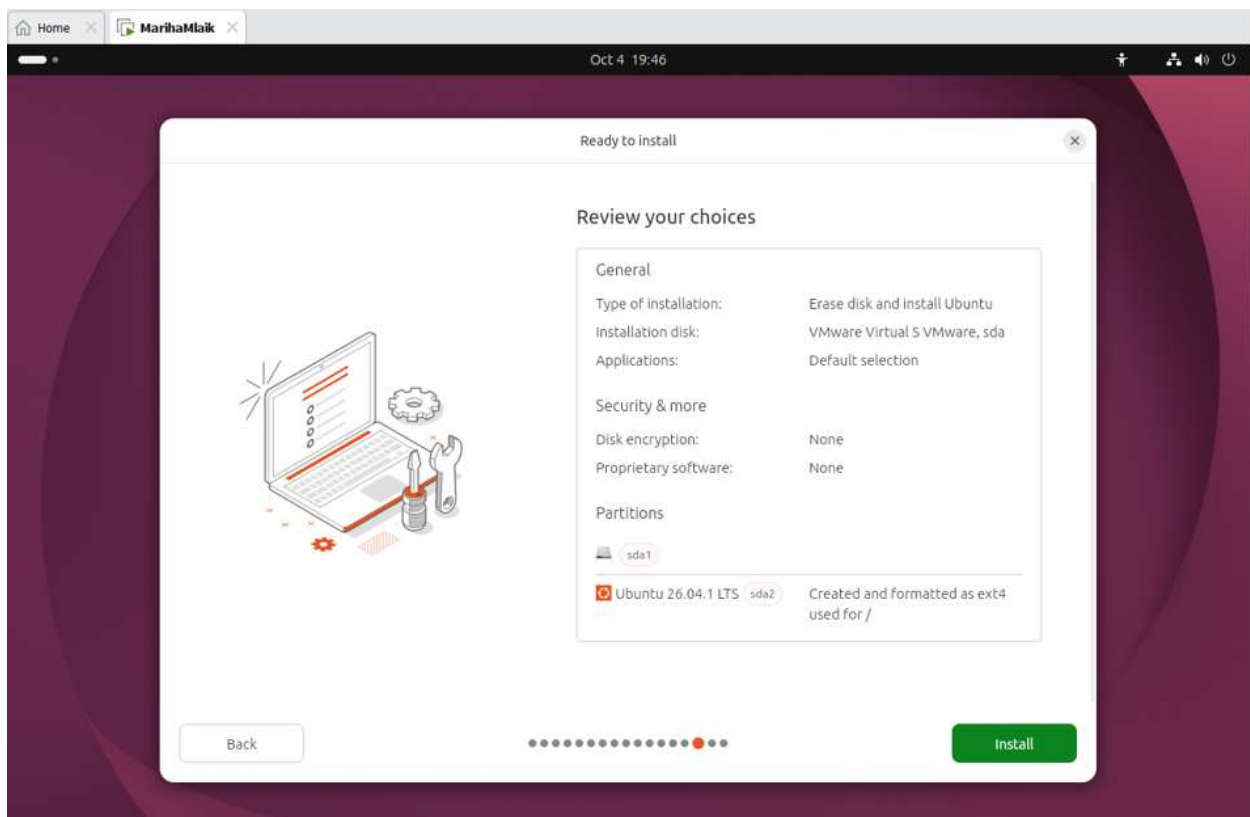
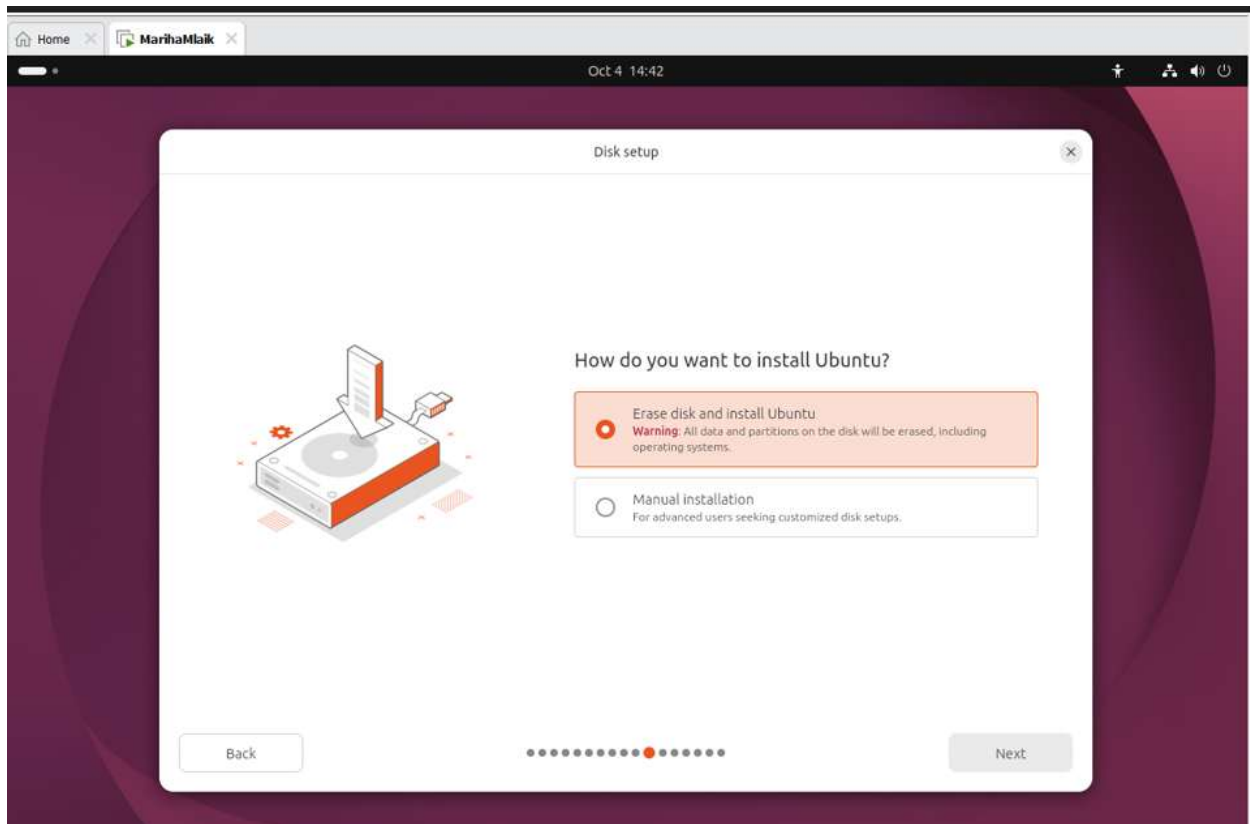
2.4. Task 7 & 8: Installation Type & Optimization

Configured standard package installation parameters and default system optimization options.



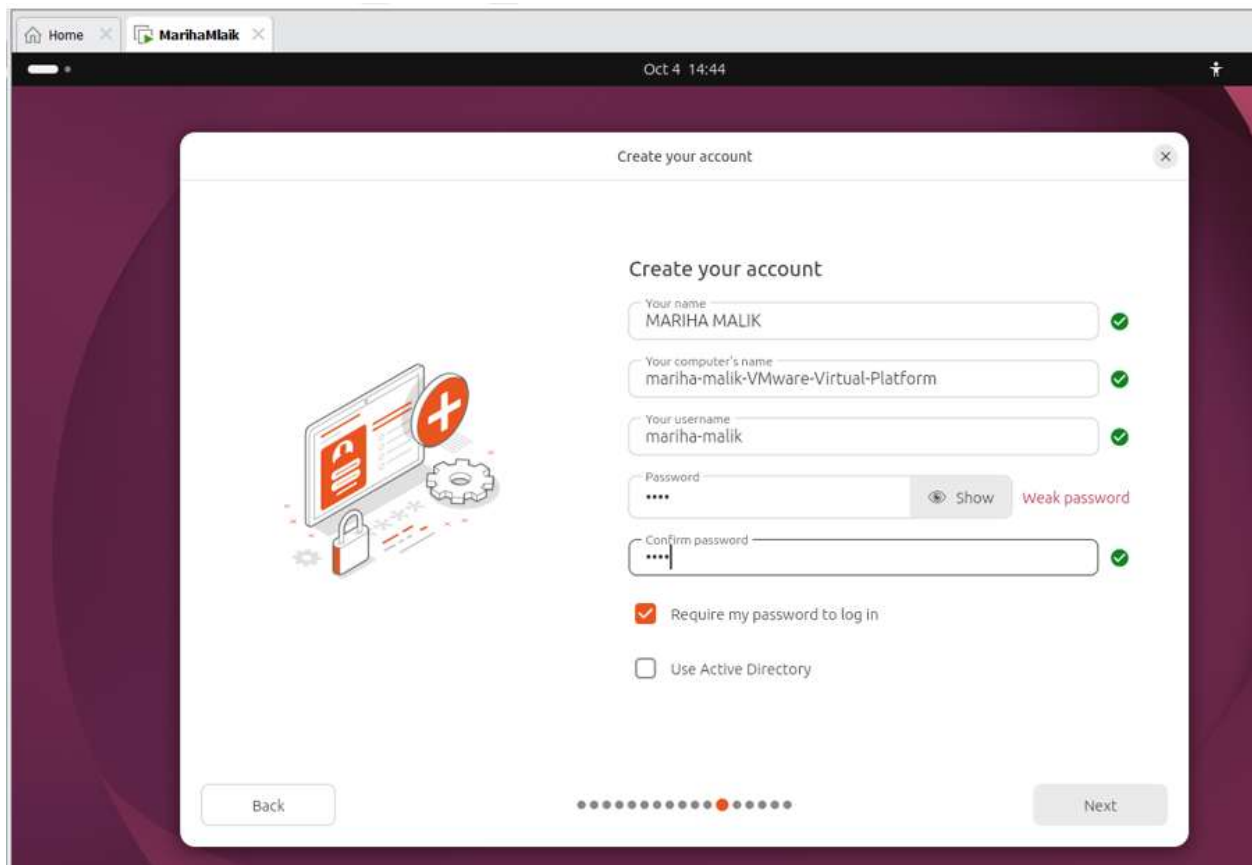
2.5. Task 9, 10, 11 & 12: Storage Layout & Disk Partitioning

Configured guided storage layout to format the virtual disk using ext4 filesystem for the root mount point and confirmed storage modifications.



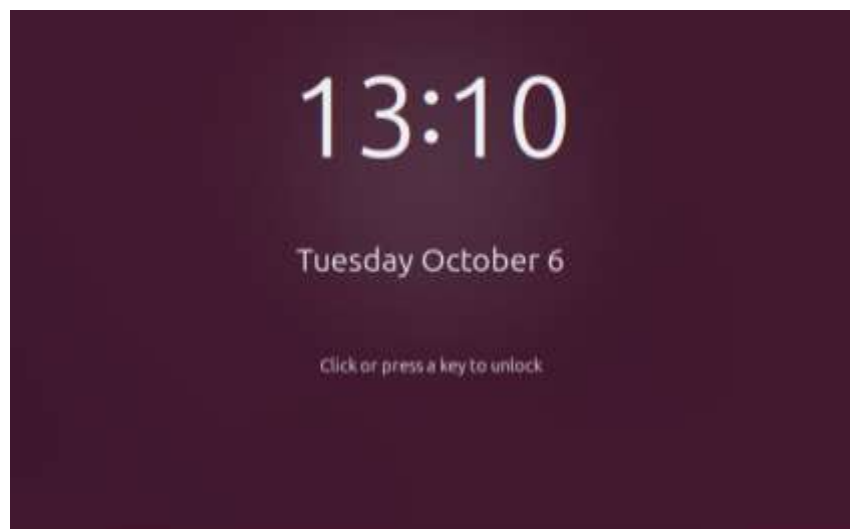
2.6. Task 13: Identity and User Profile Configuration

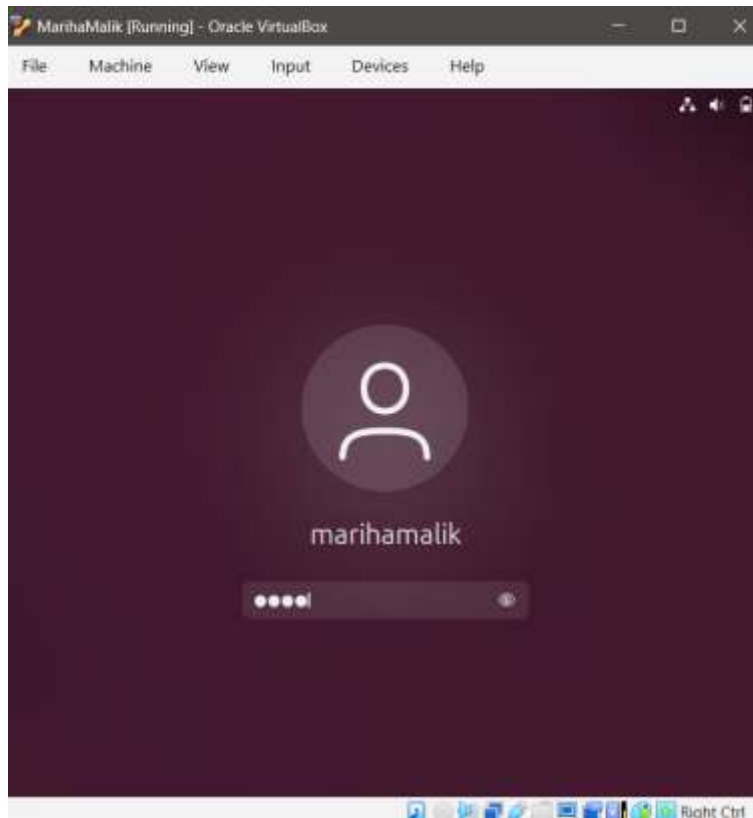
Configured primary user credentials and system hostname matching the student GitHub handle.



2.7. Task 14 & 15: Installation Completion & System Reboot

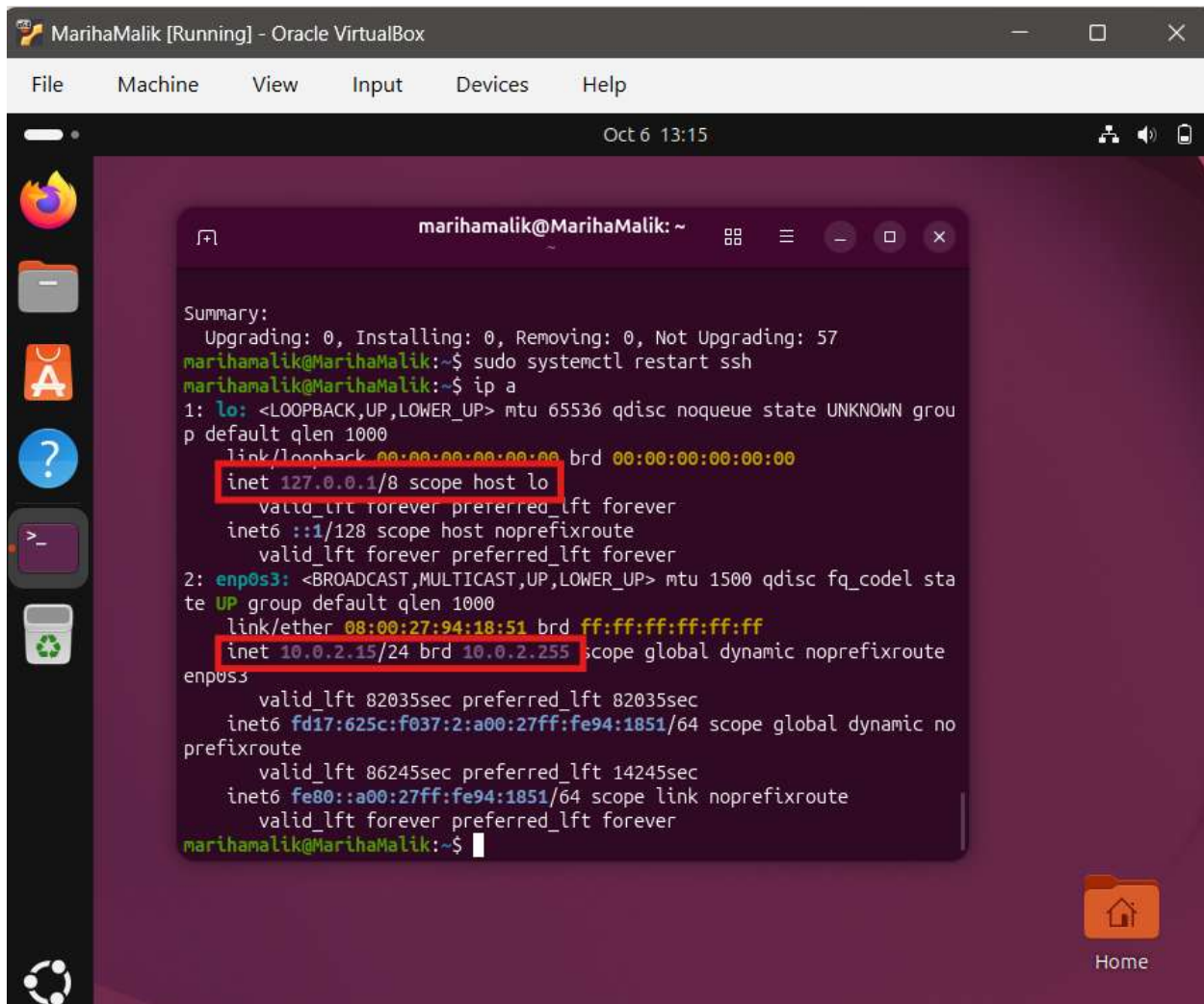
Finalized the initial operating system deployment and performed a clean reboot into the installed Linux environment.





2.8. Task 16: Local Verification and Network Configuration

Logged in locally to verify interactive shell prompt credentials and executed 'ip a' to capture network configuration details.



2.9. Task 17: Remote SSH Access from Host Machine:

Installed and enabled openssh-server, configured NAT port-forwarding (2222 to 22), and successfully established an interactive SSH session from the Windows host terminal.

```

valid_lft forever preferred_lft forever
marihamalik@MarihaMalik:~$ sudo apt install -y openssh-server
[sudo: authenticate] Password:
openssh-server is already the newest version (1:10.2p1-2ubuntu3.6).
The following packages were automatically installed and are no longer re
quired:
  linux-image-unsigned-7.0.0-30-generic
  linux-main-modules-zfs-7.0.0-30-generic
  linux-modules-7.0.0-30-generic
  linux-tools-7.0.0-30
  linux-tools-7.0.0-30-generic
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 57
marihamalik@MarihaMalik:~$

```

```
marihamalik@MarihaMalik: ~  
C:\Users\NETPOINT>ssh -p 2222 marihamalik@localhost  
The authenticity of host '[localhost]:2222 ([127.0.0.1]:2222)' can't be established.  
ED25519 key fingerprint is SHA256:AbE0B04Iw6YWhO+l4s0U4bDfGJ0i4IixU2zJ640+N0g.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '[localhost]:2222' (ED25519) to the list of known hosts.  
marihamalik@localhost's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
Expanded Security Maintenance for Applications is not enabled.  
  
57 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
marihamalik@MarihaMalik:~$
```

3. Conclusion:

The virtualized Linux environment was established, user identity verified, and secure remote connectivity confirmed via SSH.